

MUHAMMAD BASITH K

Software Engineer | Python Full Stack Developer

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Professional Summary

Software Engineer with hands-on experience in Python Full Stack Development using Django, Django REST Framework, React.js, JavaScript, and PostgreSQL. Skilled in building responsive web applications, developing RESTful APIs, implementing secure authentication, and designing scalable backend solutions. Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, SQL, Git, and modern software engineering practices, with hands-on experience developing real-world full-stack applications.

Technical Skills

Programming Languages

Python, JavaScript (ES6+), C++, SQL

Frontend

React.js, HTML5, CSS3, Bootstrap 5, Tailwind CSS

Backend

Django, Django REST Framework (DRF), REST APIs, JWT Authentication

Databases

PostgreSQL, MySQL

Developer Tools

Git, GitHub, Postman, VS Code, Render, Jupyter Notebook

Core Concepts

Object-Oriented Programming, Data Structures & Algorithms, MVC Architecture, CRUD Operations, Authentication & Authorization, API Integration, Responsive Web Design, Version Control

Professional Projects

PrepPilot – Placement Preparation Platform

Technologies: React.js, Django, Django REST Framework, PostgreSQL, Simple JWT

Links: [Repository](#) — [Live Demo](#)

- Built a full-stack placement preparation platform using React, Django, Django REST Framework, PostgreSQL, and Simple JWT for secure user authentication.
- Designed and implemented **10 custom database models** supporting study planning, placement tracking, mock interviews, calendar scheduling, and user profile management.
- Developed RESTful APIs, role-based dashboards, and company-specific mock interview generation with predefined technical and HR question sets.
- Deployed the application on **Render**, enabling public access for demonstration and testing.

City Hospital Management System

Technologies: Python, Django, PostgreSQL, Bootstrap, HTML5, CSS3, Render, Gunicorn, WhiteNoise

Links: [Repository](#) — [Live Demo](#)

- Built a role-based Hospital Management System supporting **3 user roles** (Administrator, Doctor, and

Patient) with secure authentication and authorization.

- Designed and implemented **5 custom database models** for department management, doctor profiles, appointment scheduling, user management, and medical report handling.
- Developed responsive dashboards, appointment booking workflows, medical report upload/view functionality, and automated email notifications for appointment confirmations.
- Deployed the application on **Render** using **Gunicorn** and **WhiteNoise**, configuring production-ready static and media file handling.

JobSphere – Full Stack Job Portal (Ongoing)

Technologies: React.js, Django REST Framework, Python, MySQL, Bootstrap 5, REST APIs, Git, GitHub

Status: Currently under development

- Designing and developing a full-stack job portal that enables users to browse job listings, manage profiles, and apply for jobs through a responsive web interface.
- Building secure RESTful APIs using Django REST Framework for user authentication, profile management, job posting, and application workflows.
- Implementing CRUD operations, responsive frontend components, and modular backend architecture following software engineering best practices.
- Integrating the React frontend with the Django backend using REST APIs while maintaining a clean and scalable project structure.
- Applying Git and GitHub for version control and following an API-first development approach throughout the project.

Additional Projects

Uncertainty-Aware SegFormer for Satellite Imagery (Final Year Project)

Technologies: Python, PyTorch, SegFormer (MiT-B3), Vision Transformer, Deep Learning, Computer Vision, OpenCV, Google Colab, CUDA, YAML

Links: [Repository](#)

- Developed an uncertainty-aware semantic segmentation framework by integrating Evidential Deep Learning (EDL) with the SegFormer (MiT-B3) Vision Transformer for high-resolution satellite imagery.
- Replaced conventional Softmax classification with a custom evidential head to estimate Dirichlet distributions and pixel-wise epistemic uncertainty in a single forward pass.
- Designed an optimized training pipeline using dynamic 512×512 spatial cropping, enabling efficient training of large satellite images on NVIDIA A100 GPUs while maintaining spatial alignment.
- Implemented a custom evidential loss combining Bayes Risk and Kullback–Leibler Divergence with annealing to improve model calibration and uncertainty estimation.
- Evaluated model robustness using synthetic Out-of-Distribution (OOD) corruptions and achieved approximately 1.5× faster inference than Monte Carlo Dropout while maintaining competitive segmentation accuracy.

TerraWatch – ML-Based Disaster Prediction Platform

Technologies: Python, Machine Learning, Plotly, NumPy, GIS, Raster Data Processing

Links: [Repository](#)

- Developed a geospatial analytics platform for disaster susceptibility prediction using Machine Learning and spatial datasets.

- Processed six high-resolution raster datasets and generated disaster susceptibility maps through clustering-based analysis.
- Improved usable spatial data coverage to approximately 90% by preprocessing and aligning GIS datasets.
- Built interactive 3D visualizations using Plotly to support intuitive exploration of disaster risk regions.
- Applied data engineering techniques to manage large geospatial datasets and improve prediction reliability.

Education

SRM Institute of Science and Technology, Chennai, India

Bachelor of Technology (B.Tech.) in Computer Science and Engineering

Aug 2022 – May 2026

CGPA: **7.29 / 10.0**

NM Higher Secondary School, Thirunavaya

Higher Secondary (Computer Science Mathematics)

Score: **1041 / 1200**

MES Central School, Kuttipuram

CBSE Class X

Score: **369 / 500**

Relevant Coursework

Data Structures & Algorithms

Database Management Systems

Computer Networks

Machine Learning

Object-Oriented Programming

Operating Systems

Software Engineering

Artificial Intelligence

Certifications

- AWS Academy – Machine Learning Foundations
- Oracle – Certified Foundations Associate
- Cisco – Networking Essentials
- Fortinet – Certified Associate in Cybersecurity
- Deloitte – Technology Job Simulation (Forage)

Leadership & Achievements

- Led a three-member team in developing the Uncertainty-Aware SegFormer research project for semantic segmentation of satellite imagery.
- Earned HackerRank 5 badges in C, C++, and Java along with the Problem Solving badge.
- Developed multiple end-to-end Full Stack applications using Django, Django REST Framework, React.js, and PostgreSQL.
- Experienced in building REST APIs, authentication systems, responsive web applications, and deploying Django applications.